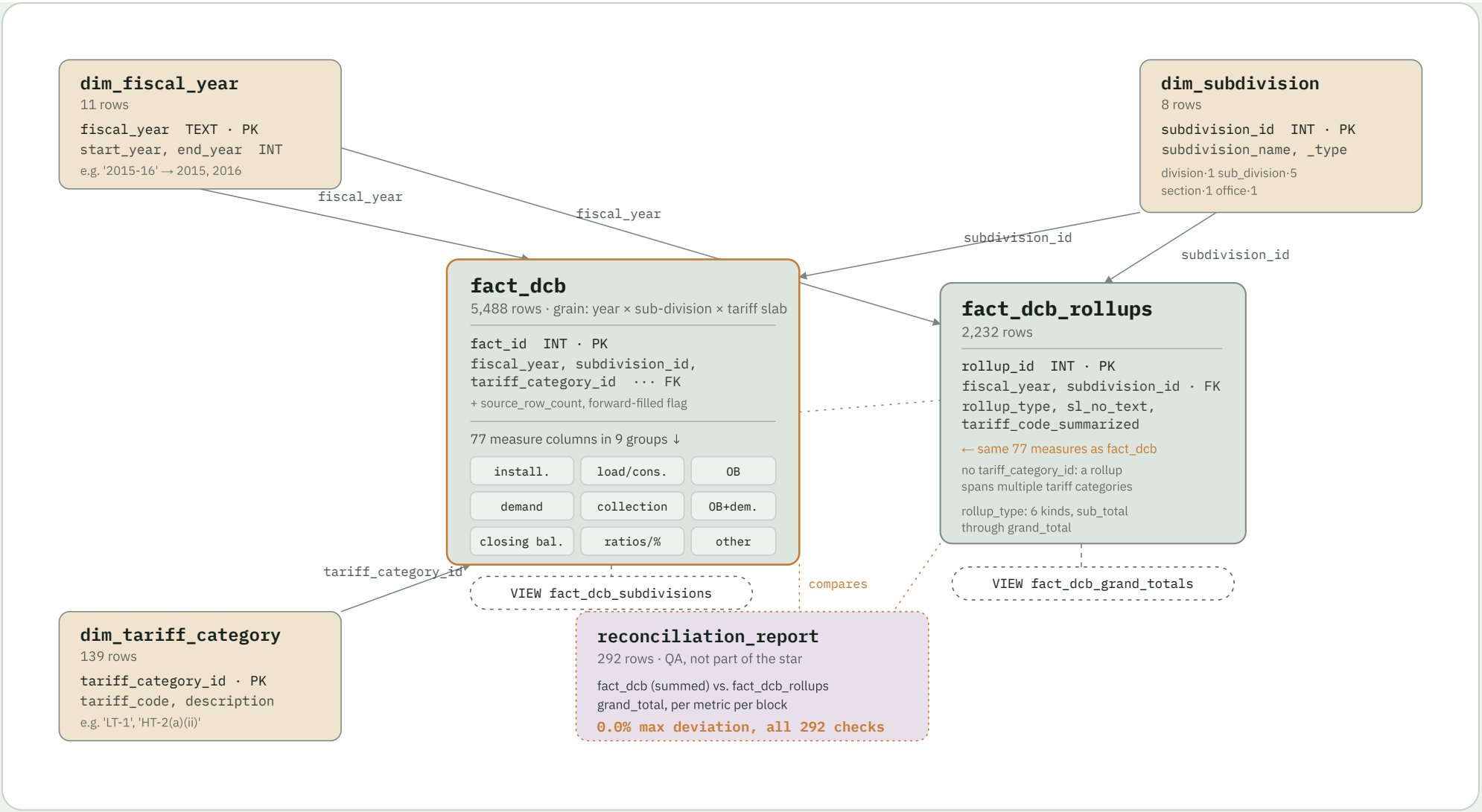
DCB-AI-APP · DATA MODEL

DCB Star Schema

Holenarasipura Division's Demand–Collection–Balance report, modeled as a star schema in `dcb_star_schema.sqlite`. Two fact tables share three dimensions — one dimension, `dim_tariff_category`, only ever joins to `fact_dcb`, since a rollup row spans more than one tariff category by definition.

11 fiscal years | **8** sub-divisions | **139** tariff categories | **5,488** fact_dcb rows | **2,232** rollup rows | **0.0%** reconciliation deviation



Solid arrows are foreign keys (many fact rows → one dimension row). The dotted copper lines show `reconciliation_report` comparing the two fact tables, not a join. Dashed pills are views, derived from the table they point into.

foreign key (many → one) ---- view, derived from ····· QA comparison, not a join

The 77 measure columns, grouped

Both fact tables carry the identical set of 77 numeric columns — the DCB arithmetic chain (Opening Balance → Demand → Collection → Closing Balance), each broken into Revenue / Interest / Tax / FAC sub-components, plus a P&G Surcharge component that only exists from fiscal year 2025-26 onward.

Installations — 6

Customer counts, including dead/non-recoverable connections.

`total_installations` `live_installations` `billed_installations` `dc_installations` `mnr_installations` `dc_mnr_installations`

Load & consumption — 7

Connected load and energy consumed.

`connected_load_kw` `connected_load_hp` `connected_load_kva` `consumption_metered` `consumption_assessed` `consumption_wheeled` `consumption_total`

Opening balance — 6

Amount owed carried in from the prior period.

`ob_revenue` `ob_interest` `ob_tax` `ob_fac` `ob_pg_surcharge` `opening_balance_total`

Demand — 11

What was billed this period.

`demand_fixed_charges` `demand_others` `demand_energy_charges` `demand_fac_standalone` `demand_pg_surcharge_standalone` `demand_revenue_subtotal` `demand_interest` `demand_tax` `demand_fac` `demand_pg_surcharge` `demand_total`

Collection — 16

What was actually paid, by component and by payment channel.

`collection_revenue_chq_cash` `collection_revenue_adjust` `collection_revenue_total` `collection_interest_chq_cash` `collection_interest_adjust` `collection_interest_total` `collection_tax_chq_cash` `collection_tax_adjust` `collection_tax_total` `collection_fac_chq_cash` `collection_fac_adjust` `collection_fac_total` `collection_pg_surcharge_chq_cash` `collection_pg_surcharge_adjust` `collection_pg_surcharge_total` `collection_grand_total`

Opening balance + demand 6

The pre-collection total each period starts from.

`ob_plus_demand_revenue` `ob_plus_demand_interest` `ob_plus_demand_tax` `ob_plus_demand_fac` `ob_plus_demand_pg_surcharge` `ob_plus_demand_total`

Closing balance 6

= Opening Balance + Demand − Collection. Becomes next period's opening balance.

`closing_balance_revenue` `closing_balance_interest` `closing_balance_tax` `closing_balance_fac` `closing_balance_pg_surcharge` `closing_balance_total`

Ratios & per-unit 14

Not additive — NULL when a fact_dcb row combines more than one source row.

`pct_live_installations` `pct_billed_installations` `pct_dc_mnr` `pct_metered_consumption` `pct_assessed_consumption` `pct_collection_efficiency_with_adj` `pct_collection_efficiency_without_adj` `pct_recovery_of_avg_cost` `ratio_arrears_to_demand` `demand_per_unit` `demand_per_unit_only` `collection_per_unit_with_adj` `collection_per_unit_without_adj` `consumption_per_installation`

Other 5

Regulatory and diagnostic figures.

`revised_arr` `security_deposit_held` `approved_avg_cost_of_supply` `diff_billed_vs_live` `diff_live_vs_total`

Generated from `DCB_Star_Schema/dcb_star_schema.sqlite` (schema DDL in `schema.sql` in the same folder). Query rules and the tariff-code glossary are in `DCB_Semantic_Layer/semantic_layer.md`. Full pipeline: raw `.xlsx` → `DCB_Segregated_By_SubDivision/` → `DCB_Staging_DB/` → `DCB_Star_Schema/` → `DCB_Semantic_Layer/` — see `HANDOFF.md`.